

EDUCATION

UNSW Sydney | Bachelor of Commerce (Finance Major) | Financial Engineering Plan **Sep 2023 – Sep 2026**

Relevant coursework: Data and Algorithms in Trading (Optiver × UNSW, **High Distinction**); Trading & Market Making; Derivatives & Risk; Financial Market Data Design and Analysis; Numerical Methods; Mathematical Economics; Data & ML.

UNSW Sydney | Master of Financial Mathematics (incoming) **Feb 2027 – Sep 2028**

UNSW International Student Award | 20% tuition-fee reduction

RESEARCH EXPERIENCE

UNSW Sydney | Research Assistant | Supervisor: Dr Xianghua Wu **Jun 2026 – Present**

- Building an automated, self-evolving research system for game-theoretic studies (new project).
- Analyzing a large-scale agent-society dataset for the AI-Agent Society & Strategic Simulation project, examining how shared-model bias, memory, information, incentives and network position shape collective outcomes.
- Drafted probe-agent, controlled-intervention and external-model counterfactual protocols; defined evaluation measures for diffusion speed, network concentration, knowledge correction and governance response.

INDUSTRY EXPERIENCE

BeixiAI (AI startup) | AI Quant Engineering Intern | Financial Event-Validation Engine **Jul 2026 – Oct 2026**

- Led the build of a LangGraph validation engine that turns research reports into excerpt-anchored claims and causal logic chains, sealed as hash-checked, replayable packages.
- Diagnosed gaps in evidence coverage, then rebuilt the RAG evidence layer with point-in-time as-of gating, verbatim span-anchored receipts and union-only vector recall.
- Designed a factor-graph variables layer under the logic chains that scores each chain by its verification record and subsequent excess returns, discounting correlated chains to guard against spurious consensus.

COMPETITIONS

NeurIPS 2026 Competition Track: Agenthon 2026 | SQA × CEWIT, Stony Brook University **2026 – Present**

Verifiable AI for quantitative finance. Entered all four tracks with a three-member team: T1 Coding (quant-finance coding agents), T2 Forecasting (reasoning-augmented time series), T3 Simulation (accelerated market simulation) and T4 Explainability (evidence-grounded prediction). In progress; final phase 13–25 Oct 2026.

Algothon 2026 | Susquehanna (SIG) × UNSW FinTech Society **Jun 2026 – Aug 2026**

Forecasting Under a Limited Effective Sample. Quant strategy developer, team abc123: **1st of 253** teams, General-round leaderboard (public data); **3rd**, Final-round technical leaderboard (500 hidden days); 69-page research report (GitHub).

- Treated 1,500 days × 51 assets as one market history with only about five independent cross-sectional units, not 76,500 independent observations, and priced selection luck (about +166 points for the best of a few hundred zero-skill variants) before comparing strategies.
- Submitted the simpler engine rather than the top scorer on the team's own leaderboard; out of sample it landed 0.4 block standard errors from the public-replay mean, 3.7 points behind 2nd place.

BigAlpha 2026 | BigQuant × Ingram Micro **2026**

Scalable Latent-Agent Market Simulator, team entry (Team Axiom): co-designed the architecture, evaluated the prototype. **2nd** in the AI Open Innovation track final; **7th** of 10 finalists (tied); review score 87, third of 16; notes on GitHub.

QUANTITATIVE PROJECTS

Signal Mosaic – Calendar-Aligned Multi-Asset & News Analytics | UNSW FINS3645 **Jul 2026 – Aug 2026**

- Engineered a reproducible 2020–2023 foundation spanning **50 US equities, 10 cryptocurrencies and 149,683 raw headlines** (2,847 exact duplicates removed); computed native-calendar returns and aligned news to the same or next equity trading day.
- Designed two descriptive diagnostics: a Coverage Lens (monthly sector-level normalized inverse-HHI of headline concentration) and a Narrative Regime Map (adjacent-year Jensen–Shannon divergence of sector vocabulary, checked with 200 equal-size subsamples).

Market Microstructure & Cross-Asset Market Making | Optiver × UNSW

May 2026 – Aug 2026

- Analyzed **1,003** timestamp-matched OB5X futures–ETF order-book observations from a single 20-minute Optibook session: negligible lead-lag within a 3-second window; return correlation rose from **0.032 at 1 second to 0.522 at 120 seconds** (only ten non-overlapping 120-second windows), consistent with an Epps effect.
- Built a cross-asset market-making system that normalized the OB5X five-stock basket, three futures and the ETF into common index units, executed depth-matched ETF–futures IOC pairs and quoted two Black-76 options with inventory-aware skew and threshold-based delta hedging.
- Ran a 1-hour Optibook simulation: simulated P&L of **+36.50** on 1,222 lots and 187 fills, with 39 arbitrage pairs matching 892 lots; maximum $|\Delta|$ was 7.51, quotes stayed two-sided throughout and the book ended flat, with no loop errors, reconnects or self-trades.

2026 AI Industry Training & Hackathon | Cognitivo × UNSW FAIC

Jul 2026

Pre-event AI Mockathon (15 Jul), team participant: a market-signal agent over RBA, ASX and AFR data with a fine-tuned domain model.

- Built the distillation set: a Qwen3.6-35B teacher labeled **630** AFR articles (90 per year, 2015–2021) from article text and the RBA rate in force, never forward returns; filtered to **518/58** train/validation rows.
- Diagnosed a silent loss-masking bug (**576** tokenization-mismatch warnings; **0/58** non-empty outputs) caused by plain-string TRL pairs; switched to chat-templated pairs with a masking check that aborts before GPU time.
- Trained the QLoRA adapter on Llama-3.1-Nemotron-Nano-8B-v1 (rank 32, alpha 64, 4-bit, 100 steps) on a GB10 shared with two live vLLM servers, then evaluated and served it with vLLM as the sentiment tool of the team's agent: held-out teacher-label agreement rose from **36/58 to 45/58**.
- After the Mockathon, audited the organizer's **60,000-row** fine-tuning corpus and 219,538 AFR records: **11,097** targets exposed 30-day future moves, the split was shuffled rather than chronological, and negative/mixed labels were absent.

Formal event (29–31 Jul): completed the 2-day AI Engineering Short Course (UNSW FAIC certificate, 30 Jul); event staff on hackathon day.

Limit-Order-Book Analytics, ETF Pricing & Market-Making Research | UNSW FINS3666

Feb 2025 – May 2025

- Reconciled **4,406 PXA trades to 24,303 eligible L1–L8 order-book states** with same-venue strict-prior matching; benchmarked four mid-price estimators and generated 34,044 rolling-100 diagnostics with zero look-ahead rows.
- Built a 5-second NDQ fair-value/iNAV model from 4,319 synchronized ETF, NQ futures and AUD/USD observations; the beta-1 specification recorded **4.30 bps MAE and 0.999 correlation** on 4,259 post-calibration rows.
- Designed a 64-session chronological minute-bar-proxy market-making study; on the 12-session historical diagnostic partition, inventory/imbalance controls cut maximum inventory from **500 to 336** and fills from 1,863 to 1,014, while diagnostic net P&L rose from **56.53 to 147.21** under identical proxy assumptions.

CURRENT RESEARCH DIRECTION

SAMS: Shared Market Representations

Sep 2026

Stateful multi-agent decision-making in limit order books (LOBs); one-page Research Brief attached. Proposed framework combining a shared market interpreter, independent participant memory and lightweight decision heads. Planned tests use game-theoretic LOB simulation and reinforcement learning to compare decision quality, strategic adaptation and end-to-end cost against matched baselines.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, Jupyter), SQL/SQLite, C++

AI & agents: LangGraph, RAG, Hugging Face transformers/PEFT/TRL (QLoRA), vLLM, FastAPI, MCP, LLM evaluation

Quantitative: order-book reconstruction, ETF fair value/iNAV, Black-76 options, market making and inventory control, point-in-time data audits

Financial engineering study: Options & Derivatives | Python & Machine Learning | C++ Programming | Advanced C++ & Modern Design

SAMS: Shared Market Representations

Stateful Multi-Agent Decision-Making in Limit Order Books

RESEARCH QUESTION

Can one shared market representation support many lightweight, stateful decision policies while reducing repeated computation and preserving strategic differences? I would test this on high-frequency limit order books (LOBs). SAMS counts as a success only if its policies match or beat independently encoded policies on fill-conditioned markouts and inventory risk under matched information, while cutting training cost and p99 decision latency; otherwise the simpler baseline stands.

RELEVANT PREPARATION

Our team's BigAlpha 2026 latent-agent market simulator (Team Axiom), whose architecture I co-designed, was 2nd in the AI Open Innovation track final; at Algothon 2026 my team placed 1st of 253 (General round) and 3rd (Final technical), choosing the final engine by effective sample size and selection-luck checks rather than raw score. I have also matched 4,406 trades to 24,303 order-book states with strict-prior timing, built an inventory-aware cross-asset market maker in Optibook and fine-tuned an 8B language model with QLoRA for an agent's sentiment tool. My current UNSW research examines shared-model bias, memory, information and incentives in agent societies.

PROPOSED ARCHITECTURE

A shared market interpreter would encode public order-flow history once per compatible information group. A compact latent interface, with learned adapters where needed, would feed lightweight participant heads. Each head would retain its own recurrent memory, inventory, outstanding orders and objective. Common inputs could be reused without pooling private state. Heads would return typed order actions, with validity and risk checks outside the model.

An event-driven game harness would enforce matching, queues and message latency, then return fills and account changes. Parallel market replicas would generate reinforcement-learning trajectories for a separate learner. Parameters and training experience could be shared; each simulated world would retain its own state. A slower interpreter and a fast incremental encoder would separate richer interpretation from event-level decisions.

INITIAL EXPERIMENTS & EVALUATION

I would first compare participant-structured prediction with direct prediction, then freeze the encoder and train one inventory-constrained market maker with recurrent PPO against fixed background policies, adding opponent-policy pools and testing on unseen opponents. I would compare SAMS with direct policies, standard parameter-sharing recurrent policies and independently encoded policies under matched information and compute budgets, and ablate memory, role conditioning and latent-update frequency.

Evaluation would cover net simulated trading results, fill-conditioned markouts and inventory risk alongside training cost and p99 decision latency, with chronological holdouts and mechanism tests for queue priority and cancel/fill races. Real-data calibration and alternative background policies would test dependence on simulator assumptions; real order-book data is not yet held.

Related work: ABIDES, MarS, LOBS5.

DEVELOPMENT PATH

In the near term, I would validate the architecture with pretrained and numerical backbones, then test LoRA and continued domain pretraining. In the longer term, I would expand market-data coverage in stages, measure data, model and system scaling separately, and test transfer by freezing the backbone and adapting new participant heads.